SOIL EROSION AND CONSERVATION · NOTE ER08

Hydrographs — Components, Separation, Unit Hydrograph and SCS Triangular Hydrograph

Course topics: SWCE3.5 L12, SWCE3.5 L13, SWCE-502 T4

Hydrograph terms

Term	Meaning
Hydrograph	Graph of flow against time — stage hydrograph (water level), discharge hydrograph, sediment hydrograph (sediment load)
Rising limb	Period of rising flow from surface and/or subsurface runoff, ending at the crest
Recession curve	Falling limb after the flood — withdrawal of water from valley, channel and subsurface storage
Retention	Precipitation that does not reach streamflow in the period — evaporation, transpiration, infiltration, percolation and temporary storage
Design flood hydrograph	Hydrograph of the limiting runoff for spillway or hydraulic design
Design flood	Largest flood a structure can pass safely; the basis of protection design
Design storm	Rainfall amount and distribution used to compute the design flood
Reservoir routing	Routing a flood through a reservoir
Base flow	Sustained dry-weather flow from groundwater, lakes, swamps or glaciers
Hydrograph separation	Splitting the hydrograph into surface runoff and groundwater components
Unit hydrograph	Storm hydrograph from unit rainfall excess of unit duration
Unit rainfall duration	Duration of the rainfall excess producing the unit hydrograph
Synthetic unit hydrograph	Unit hydrograph derived from watershed characteristics
S-curve (summation graph)	Sum of unit-hydrograph ordinates for continuous unit rainfall excess
Mass curve	Cumulative rainfall or runoff against time (mass diagram, summation graph, summation curve)

Components of runoff

- **Surface runoff:** occurs when rainfall exceeds the watershed's retention rate; flows over land to the channel, varies during the storm and stops soon after.
- **Subsurface flow (interflow, quick return flow):** infiltrated water stored at shallow depth moving downslope under gravity until a stream intercepts it.
- **Base flow:** fairly steady flow from subsurface storage, including bank storage released as water levels fall.

- **Direct runoff** = total hydrograph minus base flow — in practice surface runoff + interflow — the immediate effect of rain on streamflow.

Hydrograph analysis

- **Rising limb:** discharge rises as rain continues, controlled mainly by intensity; effective rain contributes from the start to the crest. **Crest:** the peak rate. **Recession limb:** falling flow, nearly independent of rainfall; its duration is roughly constant for a watershed.
- **Inflection point:** on the recession limb where surface inflow stops; later flow is interflow and base flow.
- **Lag time:** from the centre of gravity of effective rainfall to that of surface runoff. **Time to peak T_p:** from the start of the storm to the peak — a watershed characteristic.
- **Time base:** from the start to the end of direct runoff. The peak occurs when the whole watershed contributes at the gauging station.

Empirical relations (as in the notes): Lag time $T_l = 0.7 \cdot T_c$; Recession time $T_r = 1.67 \cdot T_p$

Separating the components

1. Plot discharge on a logarithmic scale.
2. Extend the straight recession line backward below the peak and join it to the rising limb.
3. Below the line is base flow; above it, direct runoff (surface + interflow).
4. Separate further: the lower part is interflow, the upper part direct surface runoff.

Where the boundaries lie depends on geology and watershed composition; different methods differ little in quantity.

Hydrograph type	Description
Natural	From streamflow records of gauged watersheds
Synthetic	Built from watershed and rainfall characteristics — to estimate hydrologic characteristics or modify a hypothetical natural hydrograph
Simple	One watershed unit; one rising limb, one peak, one recession; from an isolated uniform storm
Complex	Two or more simple hydrographs combined — from non-uniform intensity, storms close together, or large spatial variation. A watershed may be split into sub-watersheds and their runoff combined
Triangular	Simplest form; for sub-areas or short rainfall intervals. Adding flows at different times gives a curvilinear total hydrograph — good except near the tail
Composite	Combined hydrographs of sub-watersheds, used when the boundary is irregular or topography suggests several contributing units

Runoff volume

- Volume = area under the hydrograph (discharge × time). Direct runoff volume = area between the direct-runoff curve and the base-flow line; divide by watershed area for runoff depth.
- For rainfall events close together, first split the complex hydrograph into simple ones, then compute each direct runoff.

Unit hydrograph

- A simple natural or synthetic hydrograph for one unit depth of direct runoff (e.g. 1 cm) in a specified unit time, from an isolated rainfall excess of unit duration spread uniformly over the watershed. It is named by its excess and duration — e.g. 1.5 cm in 1 hour → a 1.5 cm–1 hour unit hydrograph.

1. **Watershed characteristics:** all physical characteristics are reflected in the hydrograph's shape.
2. **Proportionality:** for the same unit duration, ordinates are directly proportional to runoff volume.
3. **Superposition:** the hydrograph of successive excesses is obtained by overlapping the unit hydrographs of each increment.

Application: derive the unit hydrograph for 1 cm runoff and the chosen duration; find the rainfall-excess increments for the same duration; build the total hydrograph by proportionality and superposition.

> **Example (from the text)**
> Watershed 95.7 km²; storm 198 min (3.3 hr); total rainfall 5.15 cm; direct runoff 32.3 mm
> Rainfall excess = direct runoff depth = 3.23 cm. Divide the direct-runoff ordinates by 3.23 to get the unit-hydrograph ordinates.

SCS triangular hydrograph

$$\text{Lag: } T_l = 0.6 \cdot T_c$$
$$\text{Recession time: } T_r = 1.67 \cdot T_p \quad \text{Base time: } T_b = 2.67 \cdot T_p$$
$$\text{Runoff volume: } Y = \tfrac{1}{2} \cdot (2.67 \cdot T_p) \cdot q_p = 1.34 \cdot T_p \cdot q_p \rightarrow q_p = Y/(1.34 \cdot T_p)$$

The notes give the lag as 0.7 T_c in one place and 0.6 T_c here; 0.6 T_c is the SCS value.

Dimensionless hydrograph

- Plot time ratio T/T_p against discharge ratio q/q_p to get one shape independent of the watershed; the SCS triangle approximates it closely.
- **Assumptions:** T_c ≤ 6 hours for agricultural watersheds; 24-hour rain converts to 6-hour rain by a factor of 0.5–0.7; rain within the 6 hours follows a set pattern; runoff for each increment comes from the CN method; T_p = 0.7 T_c.
- **Triangular hydrograph from it:** estimate T_c; compute T_p, q_p and T_b = 2.67 T_p; plot origin → (T_p, q_p) → T_b.

Total runoff hydrograph

1. Estimate T_c.
2. Compute T_p = 0.5–0.7 T_c (or use the mean observed value).
3. Find q_p from the runoff relation.
4. Compute T/T_p for successive time steps and read q/q_p from the dimensionless hydrograph.
5. Find the runoff of each interval by the CN method.
6. Compute T_p, T_b and q_p for each rainfall increment and plot its simple hydrograph.
7. Add the ordinates to get the total runoff hydrograph.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA-NRCS National Engineering Handbook, Part 630, Chapter 16 (hydrographs). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Direct runoff equals

(a) base flow + interflow
(b) surface runoff + interflow
(c) total flow + base flow
(d) rainfall - runoff

Q2. In the SCS triangular hydrograph, the base time is

(a) 1.67 T_p
(b) 2 T_p
(c) 2.67 T_p
(d) 3 T_p

Q3. The SCS lag time is about

(a) 0.3 T_c
(b) 0.6 T_c
(c) 1.0 T_c
(d) 1.67 T_c

Q4. The unit hydrograph principle of combining successive excesses is

(a) proportionality
(b) superposition
(c) continuity
(d) linearity of time

Q5. The point on the recession limb where surface inflow ceases is the

(a) crest
(b) inflection point
(c) lag point
(d) time base

Q6. Peak of a triangular hydrograph of volume Y is

(a) Y/T_p
(b) $Y/(1.34\ T_p)$
(c) $2Y/T_b$
(d) both (b) and (c)

Q7. An S-curve is the summation of

(a) rainfall depths
(b) unit-hydrograph ordinates
(c) base flows
(d) mass curves

Answers

Q1 — (b) surface runoff + interflow
Q2 — (c) 2.67 T_p
Q3 — (b) 0.6 T_c
Q4 — (b) superposition
Q5 — (b) inflection point
Q6 — (d) both (b) and (c) — *Since $T_b = 2.67\ T_p$, $2Y/T_b = Y/(1.34\ T_p)$.*
Q7 — (b) unit-hydrograph ordinates